

• write down the matrix associated with linear transformation $T: P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R})$

defined by $T(a_0 + a_1x + a_2x^2) = a_0 + (a_0 + a_1)x + (a_2 - a_1)x^2$

w.r.t Basis $\mathcal{B} = \{1, x, x^2\}$.

• $\boxed{T(1) = 1+x}$ ✓ $T(\underset{a_0}{1} + \underset{a_1}{0}x + \underset{a_2}{0}x^2) = 1 + (1+0)x + (0-0)x^2$

• $\boxed{T(x) = x - x^2}$ ✓ $T(\underset{a_0}{0} + \underset{a_1}{1}x + \underset{a_2}{0}x^2) = 0 + (0+1)x + (0-1)x^2$

• $\boxed{T(x^2) = x^2}$ ✓ $T(\underset{a_0}{0} + \underset{a_1}{0}x + \underset{a_2}{1}x^2) = 0 + (0+0)x + (1-0)x^2$

$$M_{\mathcal{B}}(T) = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

* Any n -dimensional v.s. is isomorphic to $\mathbb{R}^n$ st. Basis

Hint Let V be n -dimensional v.s.

Let $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ be a Basis of V

$$\varphi: \mathbb{R}^n \rightarrow V$$

$$\varphi(1, 0, \dots, 0) = v_1, \quad \varphi(0, 1, 0, \dots, 0) = v_2, \quad \dots, \quad \varphi(0, \dots, 0, 1) = v_n$$

$$\varphi(x_1, x_2, \dots, x_n) = \sum_{i=1}^n x_i v_i \quad (\text{bijective l.t.}).$$

* write linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$, whose associated matrix is $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, w.r.t standard Basis of $\mathbb{R}^3$.

$$\text{st. Basis of } \mathbb{R}^3 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

$$T(1, 0, 0) = 1(1, 0, 0) + 0(0, 1, 0) + 0(0, 0, 1) = (1, 0, 0)$$

$$T(0, 1, 0) = 1(1, 0, 0) + 1(0, 1, 0) + 0(0, 0, 1) = (1, 1, 0)$$

$$\begin{aligned}
 T(1,0,0) &= 1(1,0,0) + 0(0,1,0) + 0(0,0,1) = (1,0,0) \\
 T(0,1,0) &= 1(1,0,0) + 1(0,1,0) + 0(0,0,1) = (1,1,0) \\
 T(0,0,1) &= 0(1,0,0) + 1(0,1,0) + 1(0,0,1) = (0,1,1)
 \end{aligned}$$

$$\begin{aligned}
 T(x, y, z) &= T(x(1,0,0) + y(0,1,0) + z(0,0,1)) \\
 &= xT(1,0,0) + yT(0,1,0) + zT(0,0,1) \\
 &= x(1,0,0) + y(1,1,0) + z(0,1,1)
 \end{aligned}$$

$$T(x, y, z) = (x+y, y+z, z)$$

* $\mathcal{L}(V) := \{ \text{the collection of all linear transformations from } V \rightarrow V \}$.

$\mathcal{L}(V)$ forms a v.s. over F (V is v.s. over F)

$M_n(F)$ is also v.s. over F .

* V over F , fixed an ordered basis B
 $\mathcal{L}(V) \cong M_n(F)$

◦ matrix associated with a linear transformation depends on ordered basis.

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (2x, x+y)$$

$B_1 = \text{standard basis of } \mathbb{R}^2$
 $\{(1,0), (0,1)\}$

$$T(1,0) = (2,1) \quad T(0,1) = (0,1)$$

$$M_{B_1}(T) = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

$B_2 = \{(1,1), (1,-1)\}$

$$M_{B_2}(T) = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

$$T(1,1) = (2,2) = 2(1,1) + 0(1,-1), \quad T(1,-1) = (2,0) = 1(1,1) + 1(1,-1)$$

* $M_{B_1}(T)$ and $M_{B_2}(T)$ are similar.

* Two matrices A and B are called similar if $\exists$ an invertible matrix P s.t.

* Two matrices A and B are similar if $\exists$ an invertible matrix P s.t.
 $A = P^{-1} B P$.

• P matrix is called base change matrix / Transition matrix.

• $T(x, y) = (2x, x+y)$

$B_1 = \{(1, 0), (0, 1)\}$ find P s.t.
 $M_{B_1}(T) = P^{-1} M_{B_2}(T) P$

$B_2 = \{(1, 1), (1, -1)\}$

* Express Basis vectors B_2 in L.C. of Basis vectors in B_1

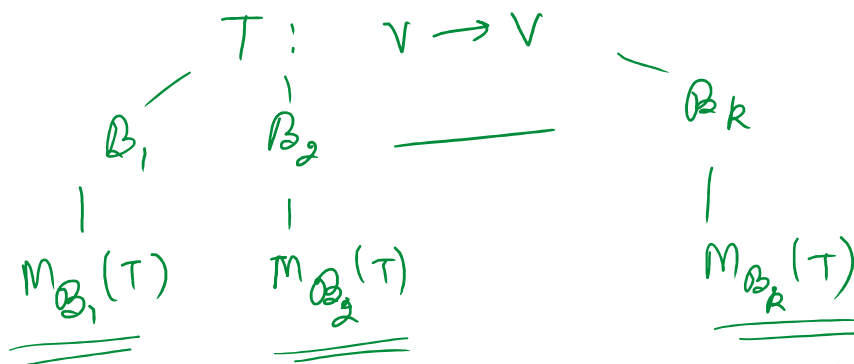
$(1, 1) = 1(1, 0) + 1(0, 1)$

$(1, -1) = 1(1, 0) + (-1)(0, 1)$

$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$P^{-1} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}$

$P^{-1} M_{B_2}(T) P = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$



• Can we find a Basis B of V w.r.t. which matrix associated with linear transformation T is a Diagonal matrix?!

* $\dim V = n$ $T: V \rightarrow V$, we want to find a Basis $B = \{v_1, v_2, \dots, v_n\}$ s.t. matrix of L.T. T w.r.t. B is a diagonal matrix (each

T w.r.t. B is a diagonal matrix

$$\begin{bmatrix} \alpha_1 & & 0 \\ & \alpha_2 & \\ 0 & & \alpha_n \end{bmatrix}$$

each
Basis
vector
is an
eigen
vector
of T .

$$T(v_1) = \alpha_1 v_1 + 0v_2 + \dots + 0v_n = \alpha_1 v_1$$

$$T(v_2) = 0v_1 + \alpha_2 v_2 + 0v_3 + \dots + 0v_n = \alpha_2 v_2$$

$$T(v_n) = 0v_1 + \dots + 0v_{n-1} + \alpha_n v_n = \alpha_n v_n$$

Eigen vector :- Let V be a vector space over F ,
let $T: V \rightarrow V$ be a linear transformation.

A $0 \neq v \in V$ is called eigen vector of T

if $T(v) = \alpha v$, for some $\alpha \in F$.

And ' α ' is called eigen value of T w.r.t Eigen vector v .

* A linear transformation $T: V \rightarrow V$ is called diagonalizable
if $\exists$ a Basis B of V s.t. $M_B(T)$ is a
diagonal matrix.

* A linear transformation $T: V \rightarrow V$ is diagonalizable
iff $\exists$ a Basis of V consisting of eigen
vectors of T .

Proof :- $\Rightarrow$ $T: V \rightarrow V$ be a diagonalizable L.T.

T.I.P. Basis B consist of eigen vectors
of T . (Done)

$\Leftarrow$ Let B be a Basis of V consisting
of eigen vectors of T .

$$T(v_i) = \alpha_i v_i$$

of eigenvectors of T .

$$B = \{v_1, v_2, \dots, v_n\}, \quad T(v_i) = \alpha_i v_i \quad \forall 1 \leq i \leq n.$$

$$M_B(T) = \begin{bmatrix} \alpha_1 & 0 & \dots & 0 \\ 0 & \alpha_2 & \dots & 0 \\ 0 & 0 & \ddots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \alpha_n \end{bmatrix}$$

α_i 's are eigenvalues of T .

$$T(v_1) = \alpha_1 v_1 = \alpha_1 v_1 + 0v_2 + \dots + 0v_n$$

$$T(v_2) = \alpha_2 v_2 = 0v_1 + \alpha_2 v_2 + 0v_3 + \dots + 0v_n$$

$$\vdots$$

$$T(v_n) = \alpha_n v_n = 0v_1 + 0v_2 + \dots + \alpha_n v_n$$